

MET CS688

WEB ANALYTICS AND MINING

ZLATKO VASILKOSKI

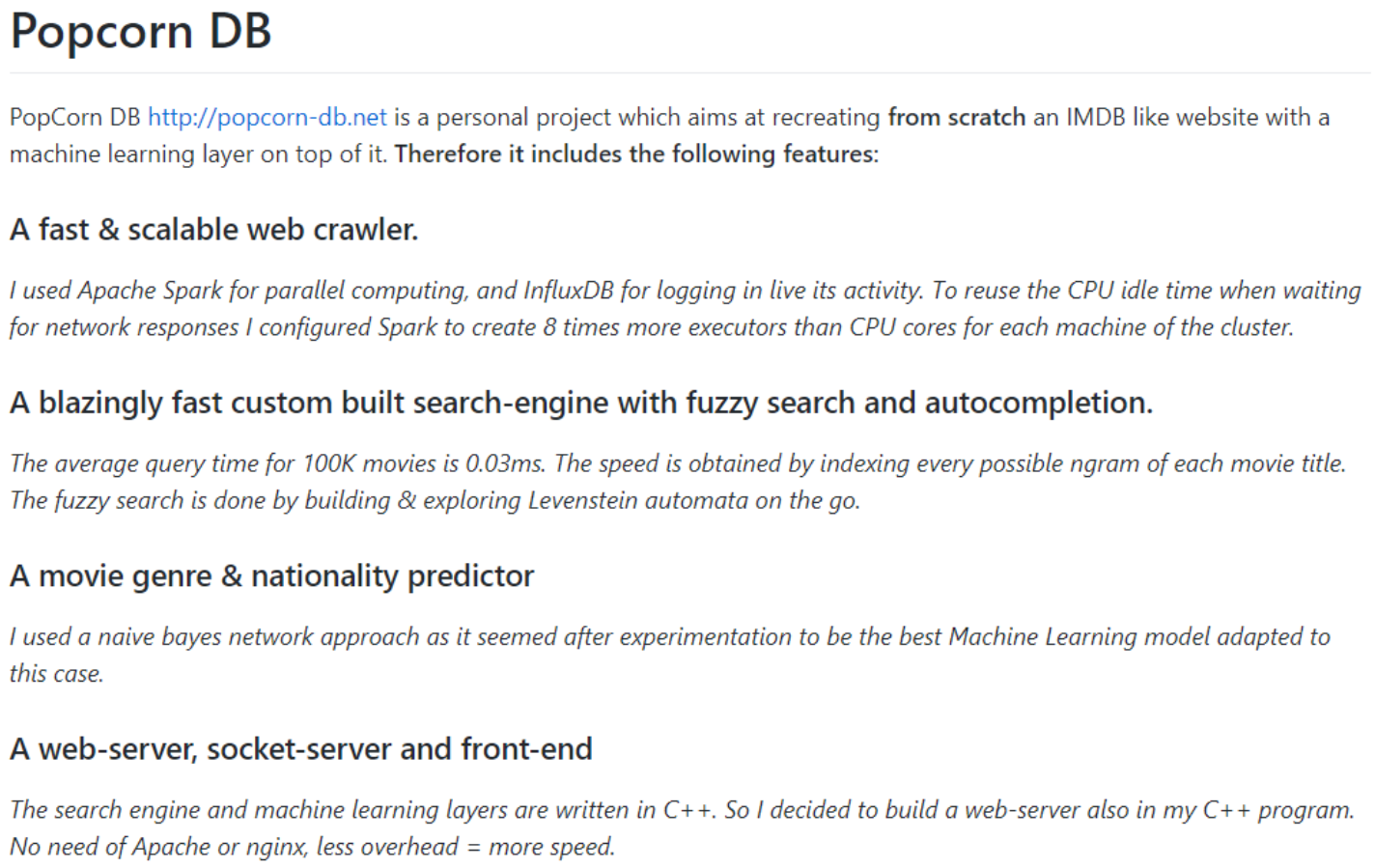
STUDENT'S TERM PROJECT EXAMPLES

Student's Term Project Examples

At the end of the semester, you will need to present a Term Project on which you would work for some time.

Here are some Term Project examples from the previous years:

- Search engine:



Popcorn DB

PopCorn DB http://popcorn-db.net is a personal project which aims at recreating **from scratch** an IMDB like website with a machine learning layer on top of it. **Therefore it includes the following features:**

A fast & scalable web crawler.

I used Apache Spark for parallel computing, and InfluxDB for logging in live its activity. To reuse the CPU idle time when waiting for network responses I configured Spark to create 8 times more executors than CPU cores for each machine of the cluster.

A blazingly fast custom built search-engine with fuzzy search and autocompletion.

The average query time for 100K movies is 0.03ms. The speed is obtained by indexing every possible ngram of each movie title. The fuzzy search is done by building & exploring Levenstein automata on the go.

A movie genre & nationality predictor

I used a naive bayes network approach as it seemed after experimentation to be the best Machine Learning model adapted to this case.

A web-server, socket-server and front-end

The search engine and machine learning layers are written in C++. So I decided to build a web-server also in my C++ program. No need of Apache or nginx, less overhead = more speed.

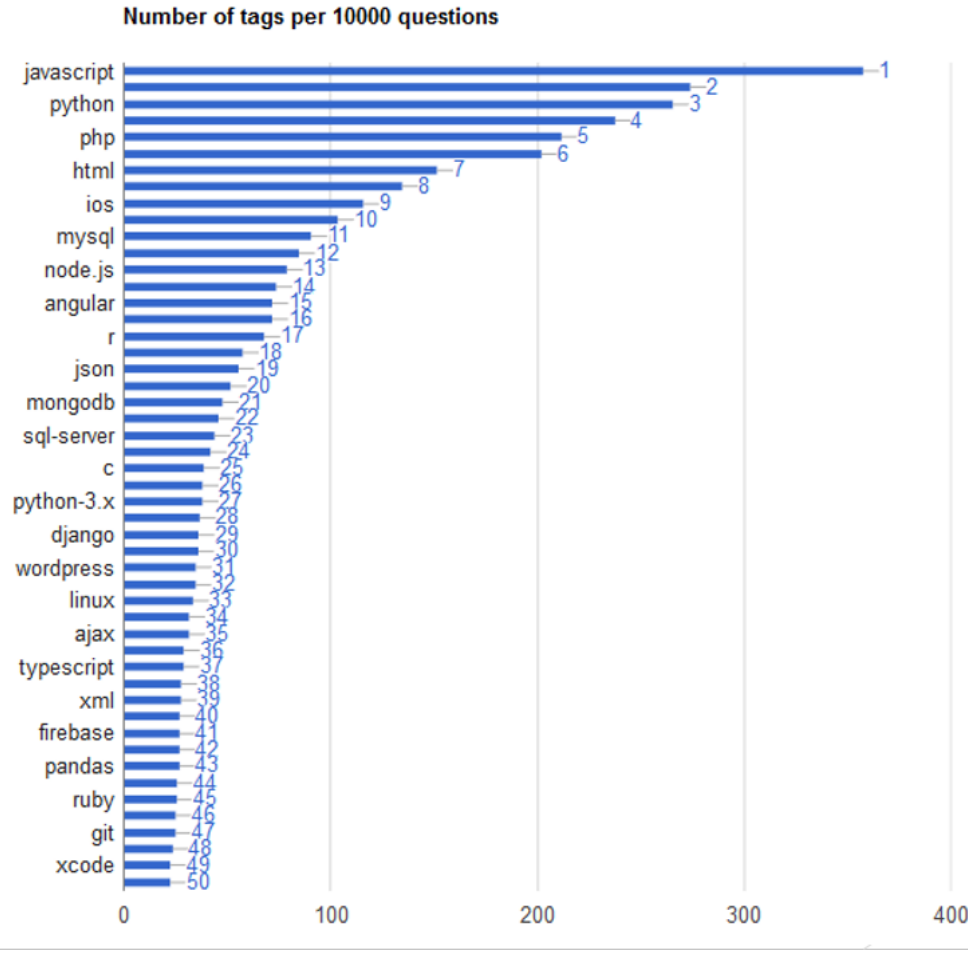
Student's Term Project Examples

Some other Term Projects Examples

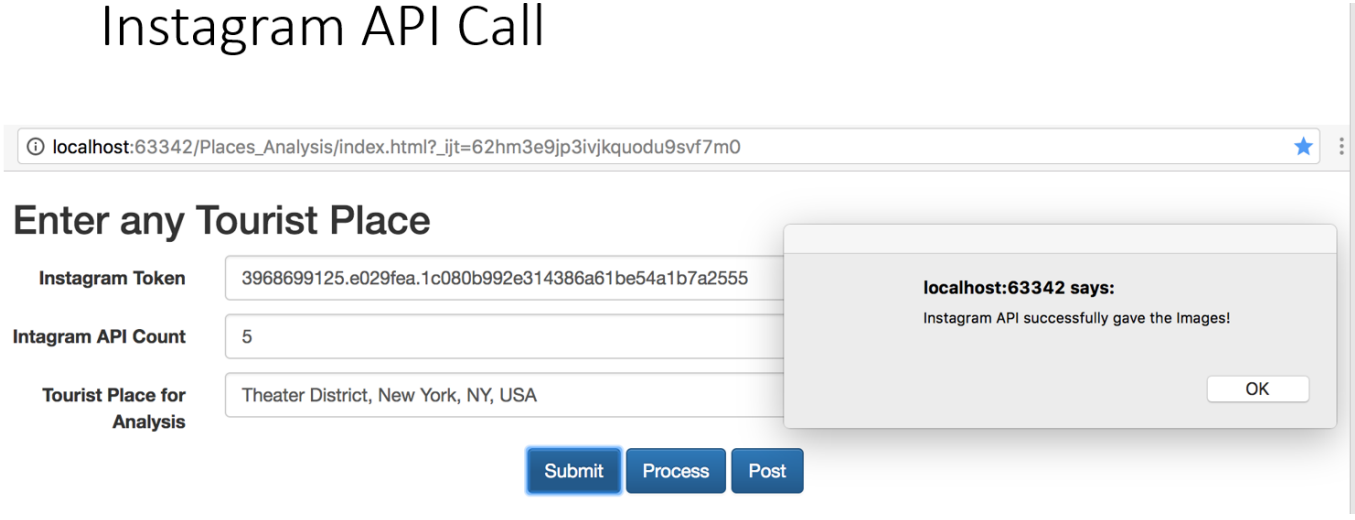
- Data mining Stackoverflow
- Places Analysis : Mining
  - Analysis on the basis of the place given by user
  - Shows the data related to the images on that location
  - Analyze the Images to get the attributes like age, gender, smile
  - Sentiment Analysis on Comments from Instagram
  - Visualization of data on Dashboard

Tags analytics

TOTAL: 36534

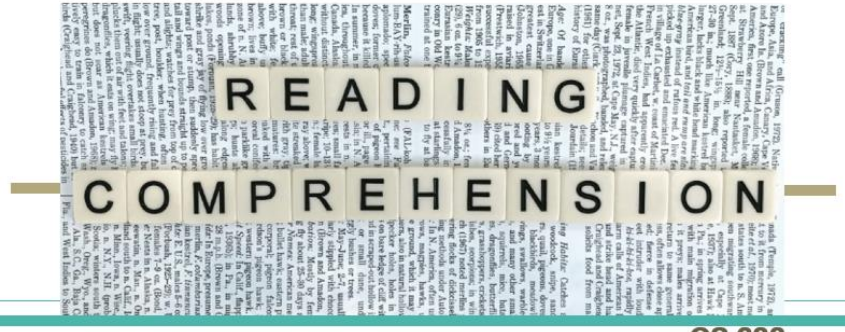


Instagram API Call



Student's Term Project Examples

BERT model-based article comprehension assistance



CS 688

Weijie Liang

Date: 08/11/2022

The data in the example is from the Chinese College English Test - CET4.

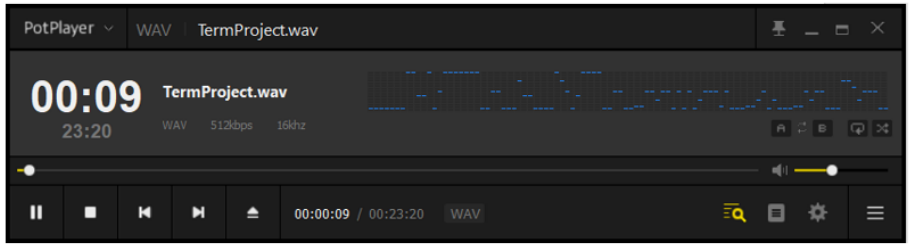
An audio file is played, and questions are asked about it.

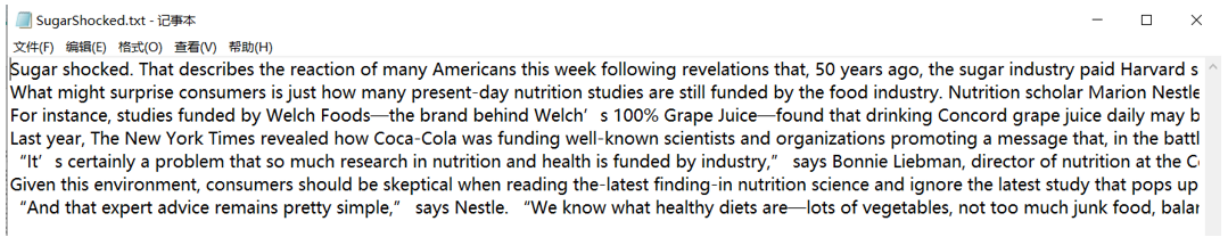
For each question there are 4 choices.

Q & A Model is created to find the most appropriate answer.

Preparing Data

For the BERT model, the easier types of questions to handle are basically listening comprehension and reading comprehension




SugarShocked.txt - 记事本

文件(F) 编辑(E) 格式(O) 查看(V) 帮助(H)

Sugar shocked. That describes the reaction of many Americans this week following revelations that, 50 years ago, the sugar industry paid Harvard s
What might surprise consumers is just how many present-day nutrition studies are still funded by the food industry. Nutrition scholar Marion Nestle
For instance, studies funded by Welch Foods—the brand behind Welch' s 100% Grape Juice—found that drinking Concord grape juice daily may b
Last year, The New York Times revealed how Coca-Cola was funding well-known scientists and organizations promoting a message that, in the battl
"It' s certainly a problem that so much research in nutrition and health is funded by industry," says Bonnie Liebman, director of nutrition at the C
Given this environment, consumers should be skeptical when reading the-latest finding-in nutrition science and ignore the latest study that pops up
"And that expert advice remains pretty simple," says Nestle. "We know what healthy diets are—lots of vegetables, not too much junk food, balar

View overall generated answer similarity

However, if we compare the generated answer with the standard answer, we can see that the generated answer is basically the same as the standard answer in terms of the general idea, but the answer given by the questioner is not directly in the original text, so the degree of similarity is not high.

46.B) They turned public attention away from the health risks of sugar to fat.

47.D) Nearly all of them serve the purpose of the funders.

48.A) Exercise is more important to good health than diet.

49.C) it rarely reuts in objective fndings.

50.D) Think twice about new nutrition research findings.

```
Q46 Question: "What did Harvard scientists do 50 years ago?"
Q46 Answer: "shifted the focus away from sugar ' s role in heart disease"
Q47 Question: "What does Marion Nestle say about present-day nutrition studies?"
Q47 Answer: "still funded by the food industry"
Q48 Question: "What did Coca-Cola-funded studies claim?"
Q48 Answer: "in the battle against weight gain , people should pay more attention to exercise and less to wha
t they eat and drink"
Q49 Question: "What does Liebman say about industry-funded research?"
Q49 Answer: "it ' s certainly a problem that so much research in nutrition and health is funded by industry
, " says bonnie liebman , director of nutrition at the center for science in the public interest . " when t
he food industry pays for research , it often gets what it pays for"
Q50 Question: "What is the author' s advice to consumer?"
```

Student's Term Project Examples

Another TP using Bert models to find the best answer to a GRE Question

Qinran Li

Can we cheat on GRE reading with BERT text similarity?

Comparing articles and choice options to get the matchest option

40 articles selected!

Questions 1 and 2 are based on this passage

The physicist Wallace Sabine pioneered the scientific study of architectural acoustics when he was asked in 1895 to fix a university lecture hall in which the echo of a speaker's words rendered them unintelligible. He found that the length of time it takes a sound's echo to decay is determined by the absorption of the sound's original energy by surrounding material. By hanging panels of sound-absorbing felt on the walls, Sabine reduced the echo enough to make the hall usable. And the data he compiled yielded a mathematical formula for the relationship between a room's echo duration, its quantity and quality of sound-absorbing materials, and its spatial volume.

EXAMPLE

The passage suggests that Sabine's work made which of the following possible for the first time?

- to make a room soundproof
- to build an auditorium out of sound-absorbing materials
- to construct an enclosed space in which sound would not echo
- to design a building to meet predetermined specifications with regard to echo duration
- to render any large room usable for public lectures and performances

Select one answer choice.

Since all questions I choose are factual details
Assumption: The most similar option to the article is correct.

GRE can be very tricky, apparently

The physicist Wallace Sabine pioneered the scientific study of architectural acoustics when he was asked in 1895 to fix a university lecture hall in which the echo of a speaker' s words rendered them unintelligible. He found that the length of time it takes a sound' s echo to decay is determined by the absorption of the sound' s original energy by surrounding material. By hanging panels of sound-absorbing felt on the walls, Sabine reduced the echo enough to make the hall usable. And the data he compiled yielded a mathematical formula for the relationship between a room' s echo duration, its quantity and quality of sound-absorbing materials, and its spatial volume.

What is your answer?

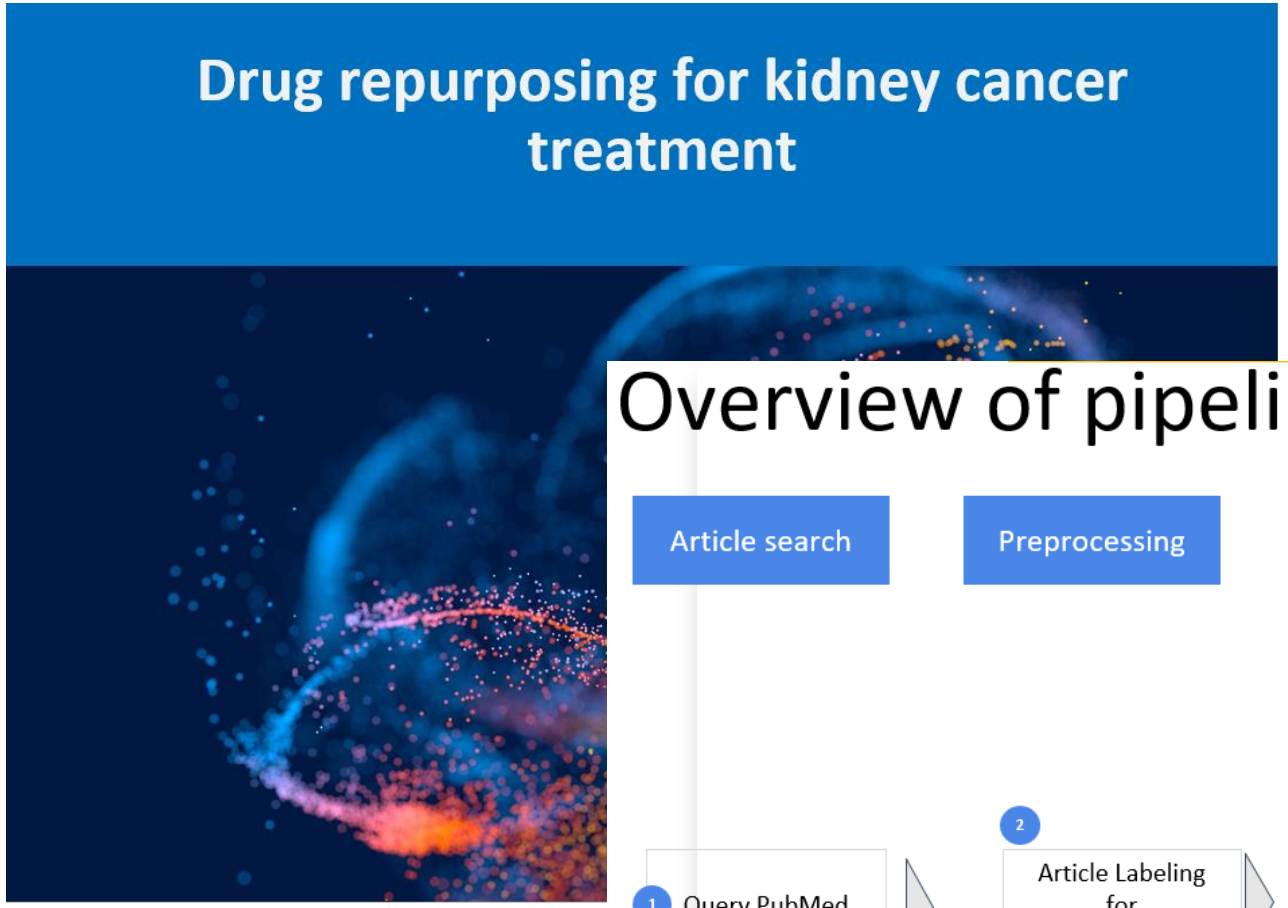


The passage suggests that Sabine's work made which of the following possible for the first time?

A. to make a room soundproof
B. to build an auditorium out of sound-absorbing materials
C. to construct an enclosed space in which sound would not echo
D. to design a building to meet predetermined specifications with regard to echo duration
E. to render any large room usable for public lectures and performances

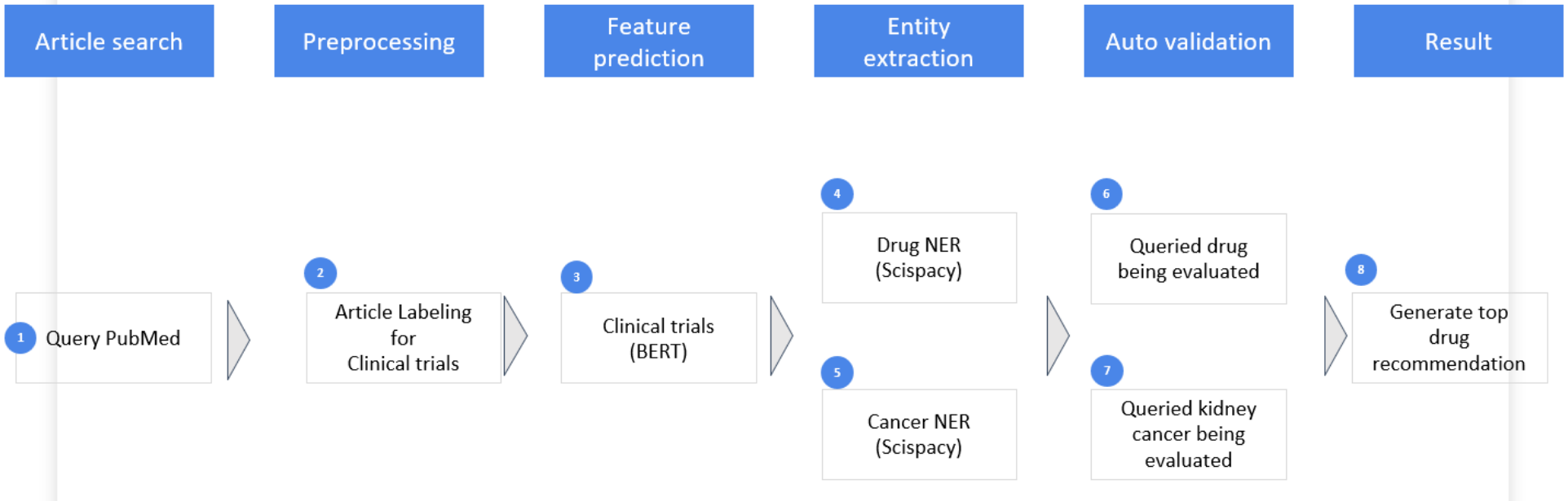
Student's Term Project Examples

Using Bert Model for drug classification



- Overview of drug repurposing pipeline
- Data retrieval using PubMed API
- Preprocessing steps
- Classification based on Clinical trial being mentioned using BERT
- Entity extraction : Scispacy NER
- EDA and Conclusion

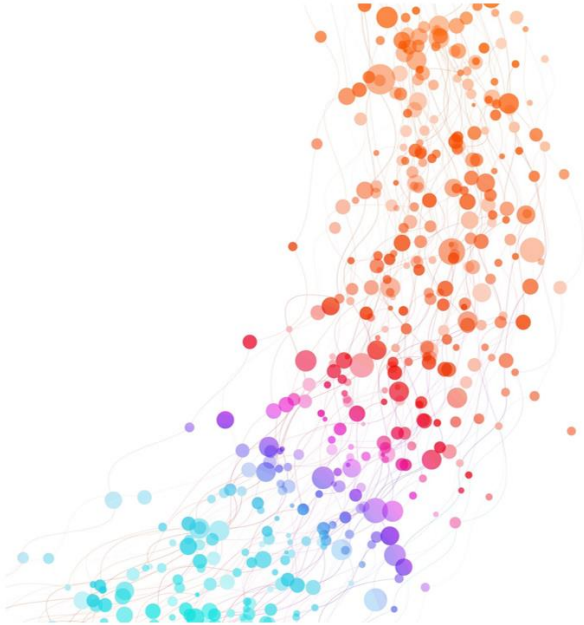
Overview of pipeline



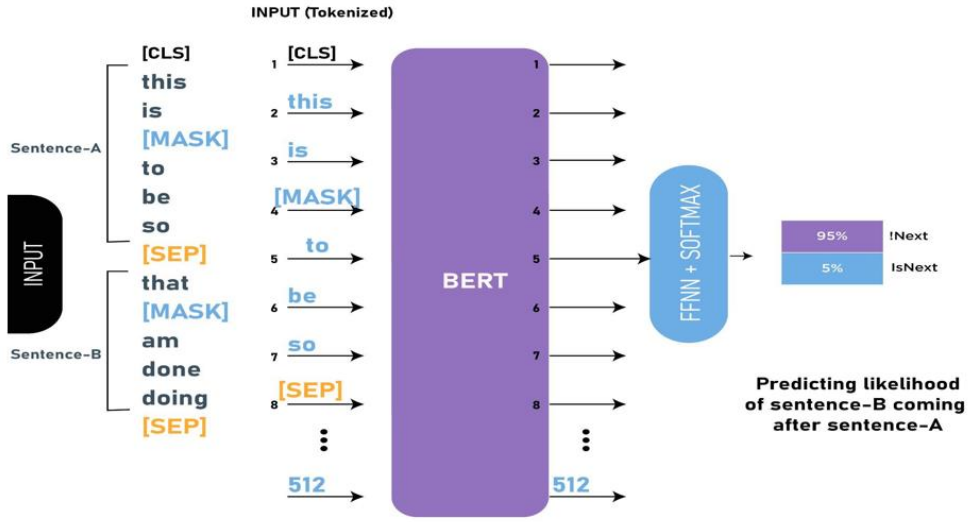
Student's Term Project Exam

FINANCIAL NEWS SENTIMENT ANALYSIS

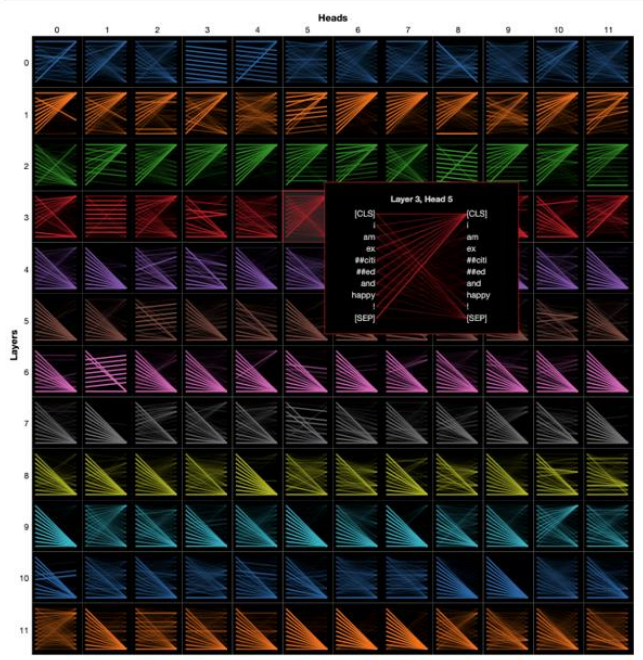
By:
Venkata Krishna Anirudh Nuti
BU ID: U39198244



BERT



BERT - Model Visualized



A

Student's TP Examples: Visual Bert, CLIP

CS 688 Project

Hateful Meme Identification using Multi-modal approach

Venkata Sandeep Yerra
Boston University

Hateful Meme Identification

- Task of detecting multimodal hate is both extremely important and particularly difficult
- Relying on just text or just images to determine whether a meme is hateful is insufficient
- Using certain types of images, text, or combinations, a meme can become a multimodal type of hate speech
- FacebookAI has released a dataset to help build systems that better understand multimodal hate speech



Text and image by themselves are not hateful .
But in combination they can become a form of hate speech

Feature Augmentation - Deepface

- Extract features from Image using Facebook Deepface (Open Source)
- DeepFace is a facial recognition system developed by Facebook researchers.
- Uses a nine-layer neural network with over 120 million connection weights to identify human faces in digital images. DeepFace was trained on four million images uploaded by Facebook users
- Regression Model
  - Age
- Classification Model
  - Emotion
  - Gender
  - Race




```
{
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     "white":66.56872034072876,
     "middle eastern":16.655877232551575,
     "latino hispanic":14.628209173679352
  },
  "dominant_race":"white"
}
```

Student's TP Examples: OpenAI/Whisper

Project Goal



- Select a podcast about the NFL (Pardon My Take)
- Transcribe the podcast of weekly NFL game recaps
- Use Sentiment Analysis to show how the Teams were talked about and predict if the Teams won or lost
- Compare the actual game results to show if teams were talked about mor favorably than expected by their record

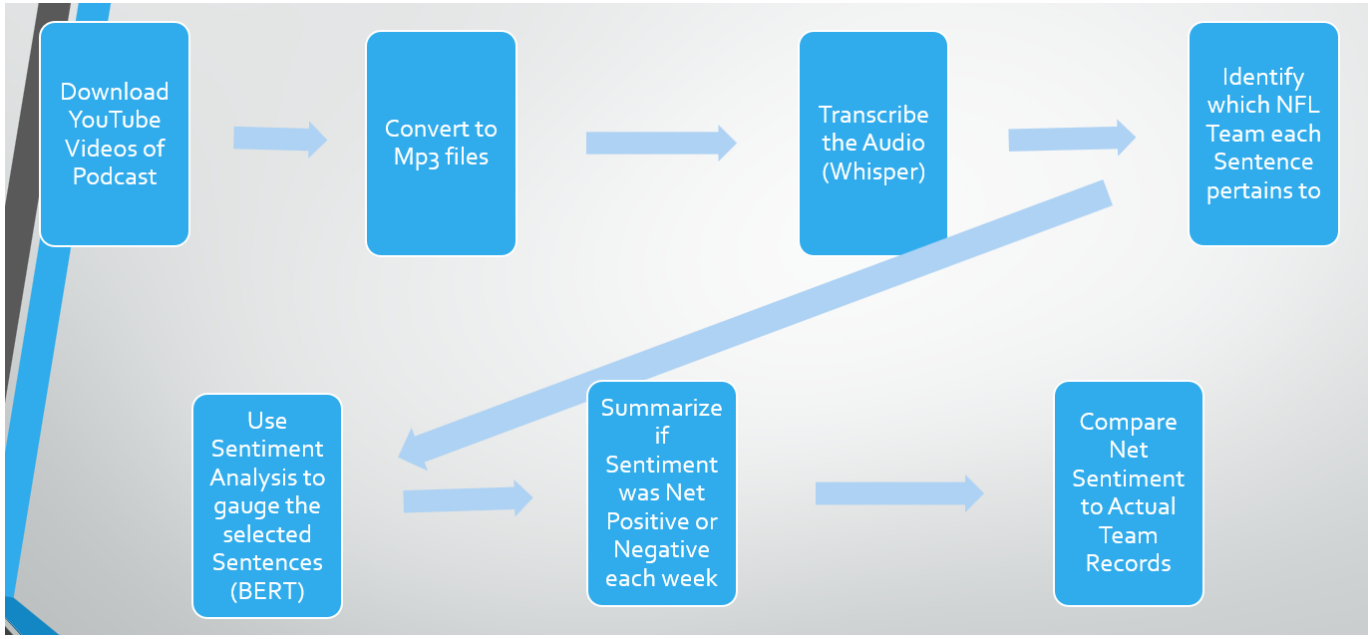
OpenAI/Whisper

Examining Bias in Sports Media using Sentiment Analysis

Brandon Bartol

CS688 Web Analytics and Mining Term Project

December 7, 2023



A

Student's TP Examples: Named Entity Recognition

Named Entity Recognition(NER) for Medical Text

Ronit Roy

User Input:

"A 57-year-old man presented to the emergency department with a 2-day history of worsening shortness of breath and chest pain. He reported no recent travel or sick contacts. His medical history was significant for hypertension, dyslipidemia, and type 2 diabetes mellitus. On examination, he was tachycardic and tachypneic, with oxygen saturation of 88% on room air. Chest radiography revealed bilateral opacities consistent with pulmonary edema. The patient was admitted to the intensive care unit for management of acute decompensated heart failure. He was started on intravenous diuretics and inotropic support with dobutamine. Over the next several days, his symptoms improved and he was discharged to home with instructions to follow up with his primary care provider in 1 week."

NER Results:

Model 1: Bidirectional LSTM Model with single LSTM layer

"A 57 Age - year Age old Age man Sex presented Nonbiological_location to the emergency Nonbiological_location department Duration with a 2 Duration -day history Sign_symptom of worsening Sign_symptom shortness Biological_structure of breath Sign_symptom and chest Sign_symptom pain History . He reported History no recent

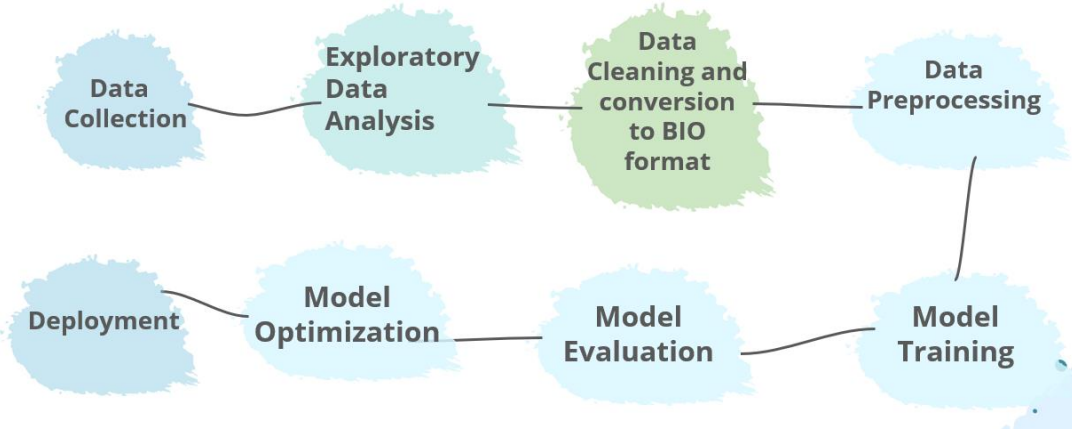
Model 2: Bidirectional LSTM Model with two LSTM layers and one Hidden Dense Layer

History history History was significant History for hypertension History , dyslipidemia History , and type History 2 Sign_symptom diabetes Detailed_description mellitus Diagnostic_procedure . On examination, he was tachycardic Detailed_description and tachypneic

Problem Statement

- Medical records are important resources in which patients' diagnosis and treatment activities in hospitals are documented.
- In recent years, many medical institutions have done significant work in archiving electronic medical records.
- Medical texts often contain complex and domain-specific entities, such as diseases, medications, symptoms, procedures, and anatomical structures.
- Many researchers strive for extracting medical knowledge from digital data, using medical knowledge to help medical professionals understand potential causes of various symptoms, and building medical decision support system

Pipeline



Student's TP Examples: Podcast to Text

01 Web Scraping
Scrape the web to download the relevant audio files.

02 Audio Transcription
Implement an Audio transcription pipeline.

03 Summarization
Implement a basic summarization pipeline for retrieving the summary of the transcribed episode.

04 Question Answering
Implement a question answering pipeline which gives us answers to question based on the transcribed episodes.

Beautiful Soup

01
Web Scraping

- Beautiful Soup is used to scrape the web and extract the audio files for each episode.
- The URLs for two popular podcasts were used.
- Additional features about each episode were extracted along with the audio file.

VOSK

02
Audio Transcription

- Vosk library is used to run a pre-trained model for the task of audio transcription.
- KaldiRecognizer is a module in Vosk which enables us to transcribe our audio files using the model of our choice from the pre-trained models included in the vosk library.

TRANSFORMERS

03
Summarization

- Hugging Face Transformers library is used to create a basic summarization pipeline.
- The pipeline makes use of the 'distilbart' model for the summarization process.

LLAMA 2

04
Q & A

- The Llama 2 7B model was used for the question answering pipeline.
- HuggingFace SentenceTransformer embeddings were used and stored in the FAISS vector store.
- The RetrievalQAChain was used to enable Question Answering capabilities while relying on a vector store to find relevant information from the document.

Prishita Kochhar

Student's TP Examples: Generative AI with LangChain

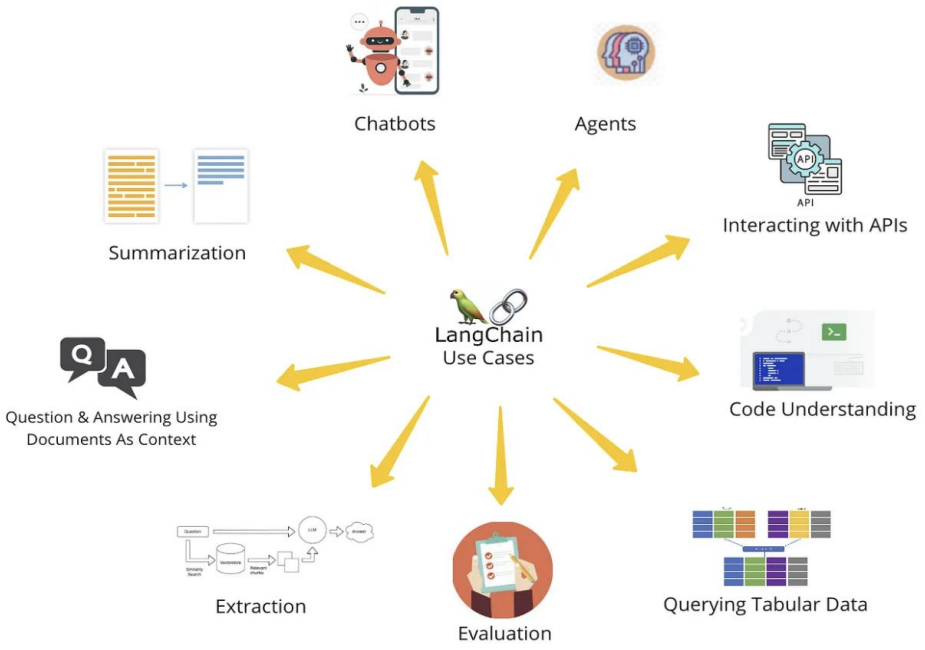
MET CS688 - Web Mining & Graph Analytics

Stop Drafting, Start Applying:AI-Powered Cover Letter Generator

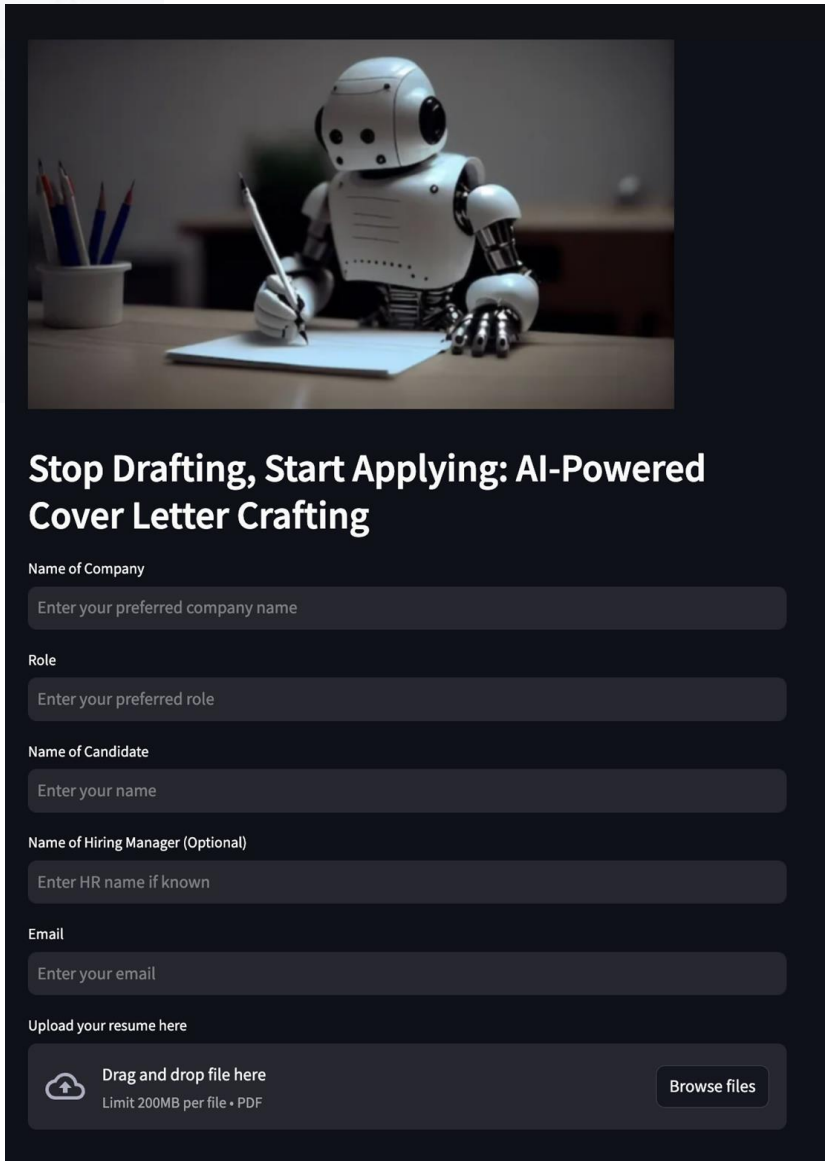
Submitted By:
Amrutha Karthikeyan
U91618654

Project Goal

- To leverage LangChain's natural language processing capabilities, powered by the GPT-3.5-turbo model, and integrate them into a user-friendly Streamlit application
- Provide a streamlined and automated option for creating customized cover letters



Streamlit App Demo



A

Student's TP Examples: Text Summary to Audio

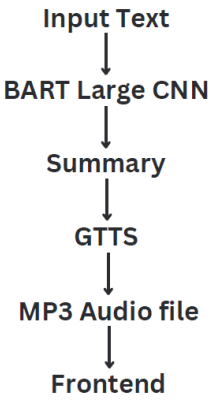
PIPELINE OVERVIEW

Input text:
Wikipedia, Journal Paper, etc

Text Summarization Transformer:
BART large by Facebook trained on CNN Daily Mail

Text-to-Speech Model:
Google's TTS (GTTS)

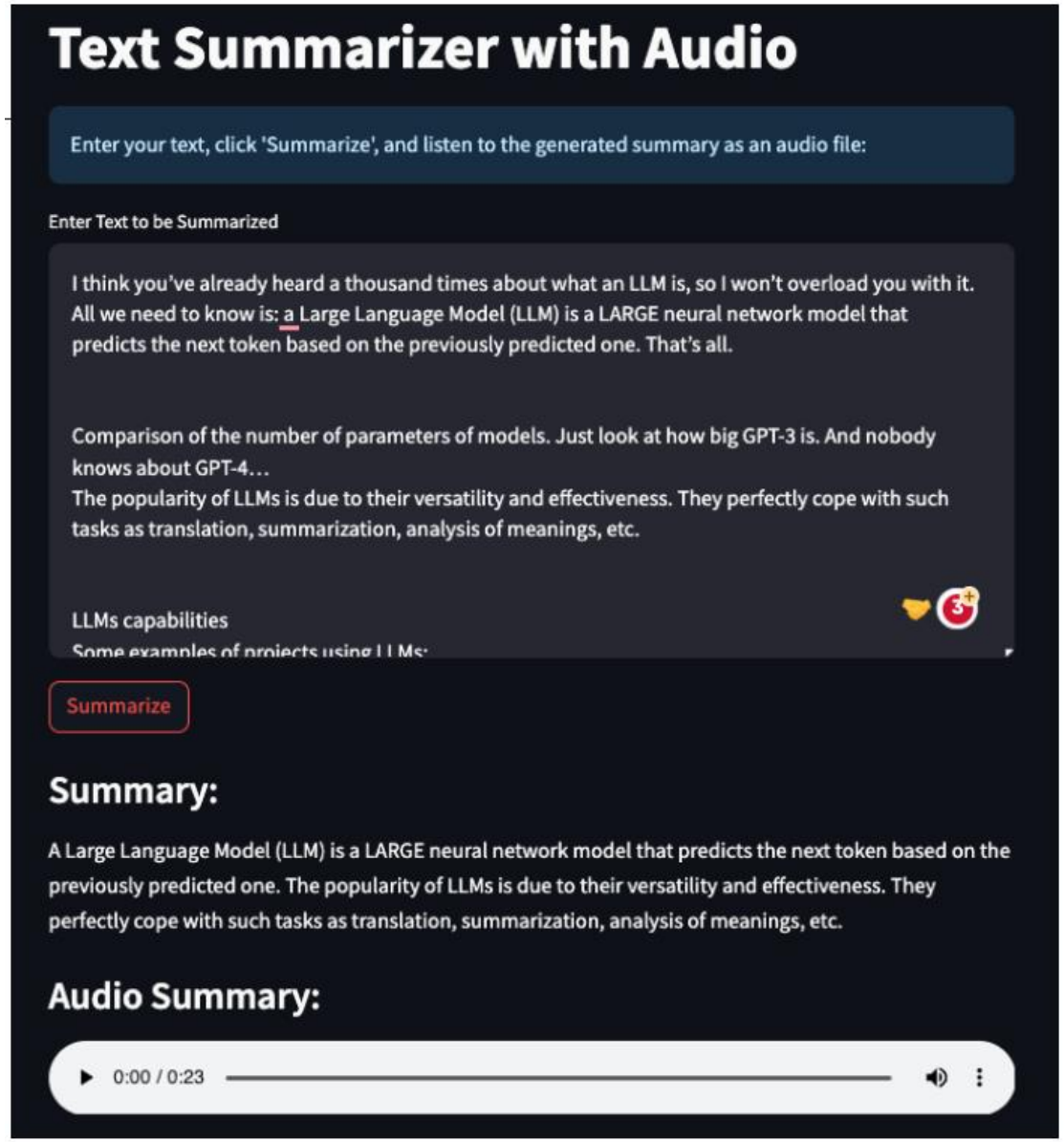
Frontend:
Streamlit





- gTTS is a Python library for text-to-speec synthesis.
- Utilizes Google's Text-to-Speech API.
- Converts text into spoken words.
- Allows customization of language, speed, and other parameters.
- Generates audio files, typically in MP3 format.

Streamlit App Demo



Keval Dave

Student's TP Examples: DL Time Series Forecasting

Multi Horizon (5) Stock Price Prediction from LOB Time-series Data (Based on Zhang's paper),
Using FI-2010 Open Source LOB Dataset



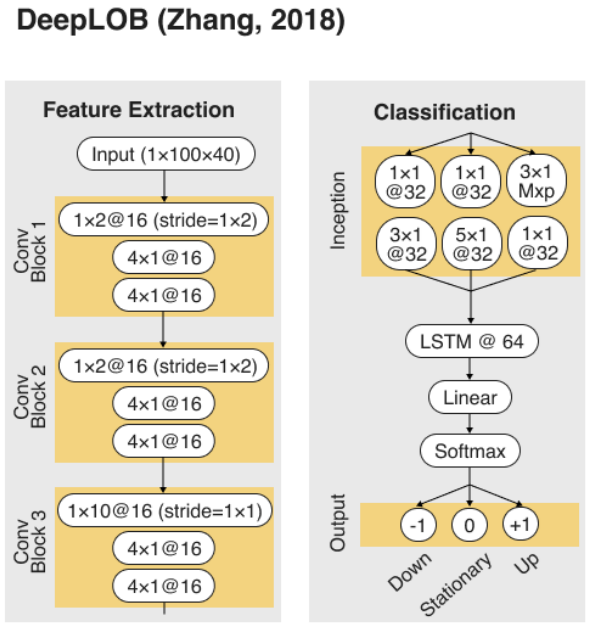


Table 4 Features Used

Feature Sets

Feature Set	Description	Details
Basic	$u_1 = \{P_i^{ask}, V_i^{ask}, P_i^{bid}, V_i^{bid}\}_{i=1}^{n}$	10(=n)-level LOB Data
Time-Insensitive	$u_2 = \{(P_i^{ask} - P_i^{bid}), (P_i^{ask} + P_i^{bid})/2\}_{i=1}^{n}$	Spread & Mid-Price
	$u_3 = \{P_n^{ask} - P_1^{ask}, P_1^{bid} - P_n^{bid}, \lvert P_{i+1}^{ask} - P_i^{ask}\rvert, \lvert P_{i+1}^{bid} - P_i^{bid}\rvert\}_{i+1}^{n}$	Price Differences
	$u_4 = \left\{\frac{1}{n}\sum_{i=1}^{n} P_i^{ask}, \frac{1}{n}\sum_{i=1}^{n} P_i^{bid}, \frac{1}{n}\sum_{i=1}^{n} V_i^{ask}, \frac{1}{n}\sum_{i=1}^{n} V_i^{bid}\right\}$	Price & Volume Means
	$u_5 = \left\{\sum_{i=1}^{n}(P_i^{ask} - P_i^{bid}), \sum_{i=1}^{n}(V_i^{ask} - V_i^{bid})\right\}$	Accumulated Differences
Time-Sensitive	$u_6 = \left\{dP_i^{ask}/dt, dP_i^{bid}/dt, dV_i^{ask}/dt, dV_i^{bid}/dt\right\}_{i=1}^{n}$	Price & Volume Derivation
	$u_7 = \left\{\lambda_{\Delta t}^1, \lambda_{\Delta t}^2, \lambda_{\Delta t}^3, \lambda_{\Delta t}^4, \lambda_{\Delta t}^5, \lambda_{\Delta t}^6\right\}$	Average Intensity per Type
	$u_8 = \left\{\mathbf{1}_{\lambda_{\Delta t}^1 > \lambda_{\Delta_T}^1}, \mathbf{1}_{\lambda_{\Delta t}^2 > \lambda_{\Delta_T}^2}, \mathbf{1}_{\lambda_{\Delta t}^3 > \lambda_{\Delta_T}^3}, \mathbf{1}_{\lambda_{\Delta t}^4 > \lambda_{\Delta_T}^4}, \mathbf{1}_{\lambda_{\Delta t}^5 > \lambda_{\Delta_T}^5}, \mathbf{1}_{\lambda_{\Delta t}^6 > \lambda_{\Delta_T}^6}\right\}$	Relative Intensity Comparison
	$u_9 = \{d\lambda^1/dt, d\lambda^2/dt, d\lambda^3/dt, d\lambda^4/dt, d\lambda^5/dt, d\lambda^6/dt\}$	Limit Activity Accelaration

Deep LOB (Pytorch) Prediction Metric:

Model Type	Normalization	Horizon	Test Accuracy
deeplob	DecPre	0	0.8342172024824779
deeplob	DecPre	1	0.7513580530034829
deeplob	DecPre	2	0.7733448952973384
deeplob	DecPre	3	0.7915263225787957
deeplob	DecPre	4	0.8018675916237871

Edward Cheng

Student's TP Examples: Reinforcement Learning

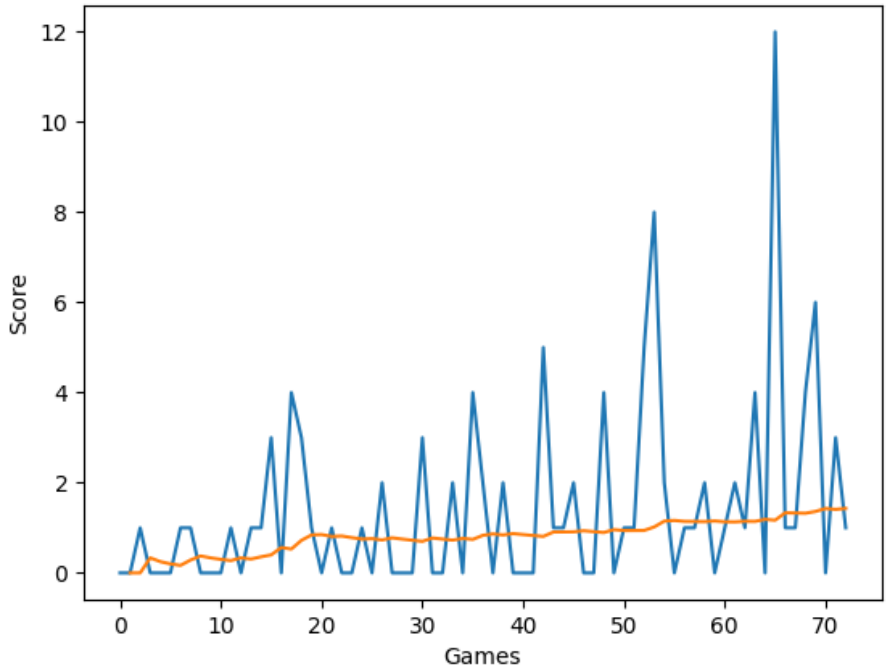
Snake Game using Reinforcement Learning – Q Learning

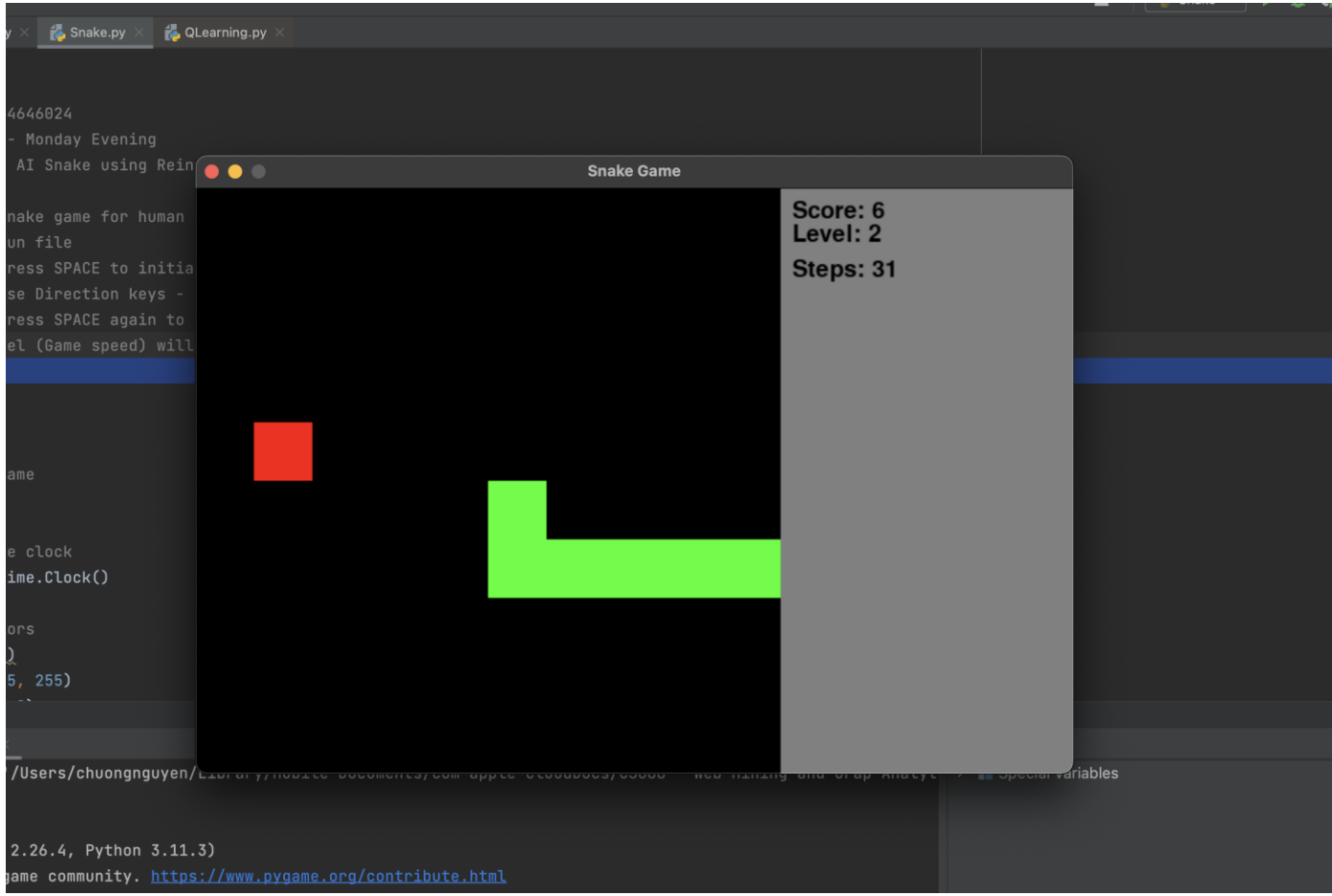
- Does not require training data
- Generate data and learn itself based on q-table

$$Q^{new}(s_t, a_t) \leftarrow (1 - \underbrace{\alpha}_{\text{learning rate}}) \cdot \underbrace{Q(s_t, a_t)}_{\text{current value}} + \underbrace{\alpha}_{\text{learning rate}} \cdot \Big(\underbrace{r_t}_{\text{reward}} + \underbrace{\gamma}_{\text{discount factor}} \cdot \underbrace{\max_a Q(s_{t+1}, a)}_{\text{estimate of optimal future value}} \Big)$$

new value (temporal difference target)

Plot of Q-Learning Model Performance





Student’s TP Examples: Neural Style Transformation

Projects

- 1. Literary Style Transformer: AI - Powered Writing Style Transformation
- 2. Neural Style Transfer App: Transform Photos into Artistic Masterpieces

Technology:

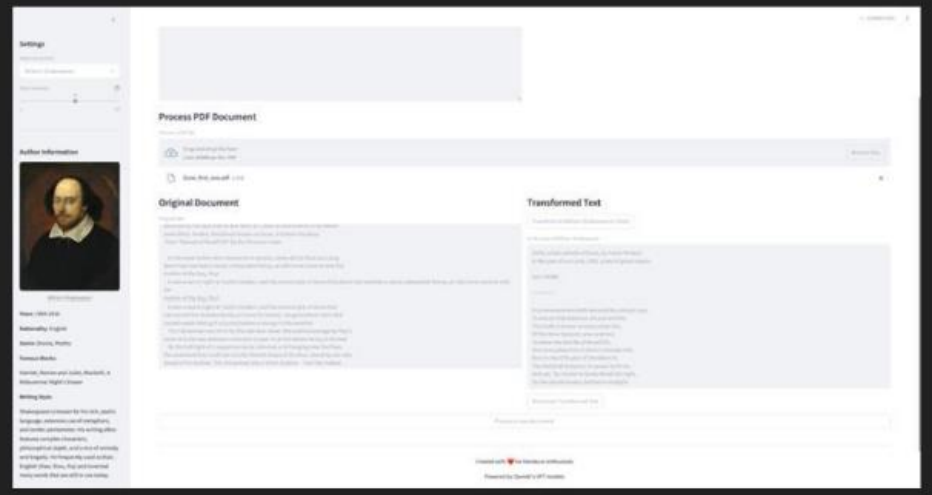
- Frontend: Streamlit
- AI/ML : OpenAI GPT-3.5 Turbo
- Langchain for PDF Processing

Solution:

- Real-time text transformation
- 13 famous authors
- Adjustable style intensity

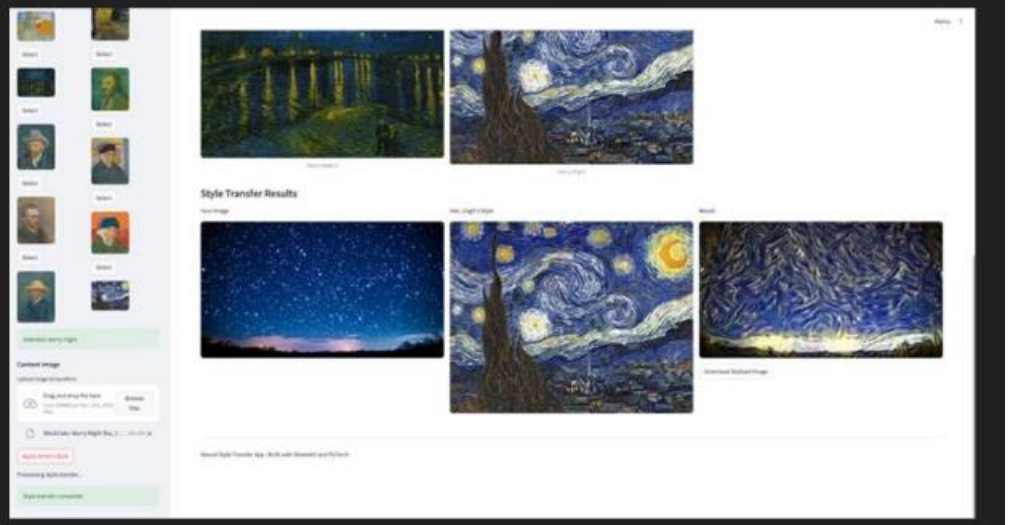
Literary Style Transformer

Demo



Neural Style Transformer

Demo



A

- A